Mini-Project (ML for Time Series) - MVA 2025/2026

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1 Introduction and contributions

The analysis of spontaneous brain activity via resting-state functional Magnetic Resonance Imaging (fMRI) is critical for uncovering the brain's intrinsic functional organization. The primary objective is to segment this activity into functional regions (atlas) to study correlations across the brain. This has significant potential for medical applications, such as developing new diagnostic and prognostic tools for impaired patients. However, this task is challenging due to the absence of a controlled experimental design (ground truth) and the noisy, high-dimensional nature of the signal.

Independent Component Analysis (ICA) has become the standard data-driven approach for this purpose, as it extracts interpretable spatial patterns that are reproducible across subjects. It operates by estimating a mixing matrix to minimize the mutual information between spatial components. However, the authors argue that ICA is limited because it lacks an explicit noise model and struggles to account for subject-to-subject variability and the spatial structure of the signal.

To address these limitations, Varoquaux et al.[1] formulate the problem within a dictionary learning framework. They propose a novel two-level hierarchical probabilistic model that learns subject-specific spatial maps and temporal components while simultaneously estimating population-level latent factors. Crucially, the authors rely on the assumption that relevant patterns are spatially sparse to reject observation noise. They introduce an efficient learning algorithm, named Multi-Subject Dictionary Learning (MSDL), based on convex optimization and validate the method on both simulated data with inter-individual variability and real fMRI data, successfully extracting a detailed population-level atlas of spontaneous activity.

In this project, we aim to apply the MSDL algorithm to novel tasks to evaluate its versatility. Specifically, we apply the method to real fMRI data to compare it against other decomposition methods in a classification task. Furthermore, we extend the application of sparse dictionary learning to video data to study the algorithm's behavior in a different spatiotemporal setting.

Regarding the implementation and distribution of work, we leveraged the `nilearn` library for data loading and processing, heavily inspiring our codebase from its documentation, but implemented our own version of the MSDL algorithm as the specific source code was not available. We introduced new experiments not present in the original article, specifically the benchmark of three decomposition methods for classification and the application to video data. Initial reading and understanding were conducted jointly; Corentin Caris implemented the fMRI analysis and the custom MSDL algorithm, while Yannaël Bossard focused on the video experiments and the Method section. All code is available at Colab fMRI and Colab video.

2 Method

2.1 Problem Statement and Generative Model

Consider a dataset comprising S subjects. For each subject $s \in \{1, \ldots, S\}$, we observe a brain signal time series matrix $Y^s \in \mathbb{R}^{n \times p}$, where n is the number of time points and p is the number of voxels. The goal is to factorize this data into k spatial latent factors and associated temporal components.

The method assumes a linear decomposition model where the observed data Y^s is generated by a product of subject-specific time series $U^s \in \mathbb{R}^{n \times k}$ and subject-specific spatial maps $V^s \in \mathbb{R}^{p \times k}$:

$$Y^s = U^s (V^s)^T + E^s, \quad \text{with} \quad E^s \sim \mathcal{N}(0, \sigma I) \quad \text{and} \quad U^s \sim \mathcal{N}(0, \Sigma_U) \tag{1}$$

where E^s represents the observation noise, assumed to be isotropic Gaussian with variance σ.

To account for inter-subject variability while maintaining correspondence across the population, the model introduces a hierarchical constraint, by making the subject-specific maps V^s as deviations from a shared population-level atlas $V \in \mathbb{R}^{p \times k}$:

$$V^s = V + F^s, \quad \text{with} \quad F^s \sim \mathcal{N}(0, \zeta I) \tag{2}$$

F^s enables to capture the subject-specific variability (e.g., local topographic differences) with variance ζ. Finally, a prior distribution is placed on the group-level maps V to enforce specific properties such as sparsity and spatial coherence: $\mathcal{P}(V) \propto \exp(-\xi \Omega(V))$, where $\Omega(V)$ is a regularization penalty defined below 2.3.

2.2 Optimization

To estimate the parameters, the framework employs a Maximum A Posteriori (MAP) estimation, so-called MSDL, for estimation from the data of Eq. 1 and 2. This results in maximizing the log-likelihood of the data combined with the penalization terms. The resulting optimization problem is to find the dictionaries U^s, V^s and the atlas V that minimize the following energy function $\mathcal{E}$:

$$\mathcal{E}(U^s, V^s, V) = \sum_{s=1}^{S} \frac{1}{2} \|Y^s - U^s (V^s)^T\|_{Fro}^2 + \sum_{s=1}^{S} \frac{1}{2} \mu \|V^s - V\|_{Fro}^2 + \lambda \Omega(V) \tag{3}$$

The first term ensures the learned subject maps V^s and time series U^s accurately reconstruct the original fMRI data. The second term keeps subject-specific maps V^s close to the group-level atlas V. The third term ensures spatially sparse but structured group atlas V. The hyperparameters are defined as $\mu = \frac{\sigma}{\zeta}$ (controlling the deviation of subjects from the atlas) and $\lambda = \frac{\sigma}{\xi}$ (controlling the sparsity of the atlas). To prevent arbitrary scaling of the components (i.e., avoiding a degenerate solution with a shrunken V compensated by a very large U^s), the scale of U^s is fixed such that $\|u_l^s\|_2 \leq 1$ (u_l^s for the l-th column).

2.3 Regularization: The Smooth-Lasso

A key component of the method is the choice of the penalty $\Omega(V)$. Standard ℓ_1 penalties (Lasso) induce sparsity but fail to account for the spatial structure of brain images, often resulting in scattered, "salt-and-pepper" noise, where all voxels are treated independently. To promote spatially contiguous regions, the authors propose using the Smooth-Lasso (SL) penalty:

$$\Omega_{SL}(V) = \sum_{l=1}^{k} \left(\|v_l\|_1 + \frac{1}{2} v_l^T \mathbf{L} v_l \right) \tag{4}$$

where v_l is the l-th spatial map (column of V) and $\mathbf{L}$ is the Laplacian operator defined on the 3D voxel grid, penalizing sharp jumps in values. This SL penalty enforces both sparsity (via the ℓ_1 norm) and spatial smoothness (via the ℓ_2 penalty on the gradient).

2.4 Algorithm

The objective function in Eq. (3) is minimized via a block coordinate descent strategy that scales linearly with the number of subjects S. The procedure iterates through three steps: (1) updating time series U^s via constrained least-squares; (2) estimating subject maps V^s via ridge regression to balance data fit and atlas proximity; and (3) updating the group atlas V by applying the proximal operator of Ω to the average of subject maps. For the Smooth-Lasso, this proximal step is computed efficiently using the FISTA algorithm.

Since the optimization problem is non-convex, the final solution relies on good initialization. To this end, the group maps V are initialized using the FastICA algorithm. In fact, ICA extracts super-Gaussian signals, which serve as a proxy for sparsity. To ensure the relevance of the initial maps, the method runs ICA with an increasing order and selects k components that exhibit positive kurtosis.

2.5 Model selection and Parameters tuning

To ensure the model generalizes well without manual tuning, the authors propose data-driven strategies for setting the hyperparameters. The subject-level penalization μ (controlling inter-subject variability) is estimated by comparing the variance of residuals from a subject-specific PCA against the residuals of a group-level PCA on concatenated data. The population-level penalization λ (controlling sparsity) and the model order k (number of regions) are selected via 3-fold stratified cross-validation. The goal is to maximize the likelihood of left-out data, ensuring the learned atlas captures true functional structure rather than noise. While higher model orders generally increase likelihood, they found that it can lead to saturation.

3 Data

3.1 MSDL Atlas and fmri analysis

The analysis presented in this study utilizes two primary datasets: the MSDL atlas and the Development fMRI dataset.

The MSDL atlas (Fig 1) is a probabilistic framework consisting of 39 functional regions of interest (ROIs) distributed across several key brain networks, including the default mode (DMN), auditory, motor, and visual systems. Unlike deterministic parcellations, the MSDL atlas provides continuous probabilistic maps, allowing for overlapping representations of functional nodes.

Complementary to this, the developmental dataset comprises movie-watching fMRI scans from 50 subjects, categorized into two age groups: children and adults. Preprocessing of the functional data involves standardization (z-scoring), detrending, and band-pass filtering within a frequency range of 0.01 to 0.1 Hz to isolate relevant physiological signals. Additionally, potential artifacts were addressed through confound regression, utilizing provided movement and physiological parameters. Signal extraction was performed using both the probabilistic atlas and region extraction techniques to derive subject-specific time series, which served as the foundation for constructing functional connectomes via various connectivity measures such as correlation, partial correlation, and tangent space embedding.

3.2 Video Data for MSDL Evaluation

To evaluate the MSDL framework outside its native fMRI context, we curated a suite of video datasets. Mathematically, video analysis parallels fMRI analysis as both rely on decomposing spatiotemporal volumes (2D pixels + time for video vs 3D voxels + time for fMRI) using the linear model $Y \approx UV^T$. We designed three synthetic datasets (100 frames, 50×50 pixels) and selected one open-access benchmark to diagnose specific model behaviors. All used videos are humanly interpretable.

1. **Dataset A (Model Validation):** We generated fixed spatial blobs with correlated temporal activation. This dataset strictly adheres to the MSDL generative model, which assumes subject-specific maps are generated from fixed population maps and time series follow a multivariate Gaussian distribution $U^s \sim \mathcal{N}(0, \Sigma_U)$.
2. **Dataset B (Apparent Motion):** To test temporal recovery, we simulated a sequence of five fixed blobs activating in order $(A \rightarrow B \rightarrow C)$. While the spatial maps remain fixed, the deterministic periodicity of the signal challenges the stochastic Gaussian prior of the model.
3. **Dataset C (Continuous Motion):** We generated a single blob moving continuously in a spiral. This represents a fundamental violation of the model: the "spatial map" changes at every frame, contradicting the assumption of fixed latent factors. Furthermore, the moving object introduces a heavy positive skew, violating the zero-mean Gaussian residual assumption.
4. **Real-World Data (BMC):** We utilized the "121.mp4" video from the *Background Models Challenge*. The video depicts a street roundabout with cars (sparse moving foreground) against a static background. Although MSDL is designed for extracting dictionary/components that are consistent across time and across subjects, as it is the case here, such videos violates the model assumptions. The static background is removed by the centering step, while the cars represent sparse, non-stationary outliers that defy the Gaussian assumptions of the temporal loadings as much as the fixed maps assumption.

4 Results

4.1 From dictionary learning to functional connectomes

We conducted experiments to evaluate the transition from spatial brain decomposition to time-series connectivity analysis.

Spatial Map Derivation and Implementation: We evaluated three decomposition methods: CanICA, standard Dictionary Learning, and our hierarchical MSDL implementation. As seen with the pretrained ROIs, the MSDL approach, which models subject-specific variations ($\mathbf{V}^s$) around a group atlas ($\mathbf{V}$), successfully produced localized regions (Fig 2). We observed that the Smooth-Lasso penalty was critical; without spatial smoothness (λ_2), the derived time-series sensors were too fragmented, leading to noisy temporal signals.

Connectivity Modeling and Classification: Extracted time series were used to build functional connectomes using three metrics. Quantitatively, the **Tangent Space** (Fig 3) measure seemed to outperform both Correlation and Partial Correlation, achieving classification accuracies for the "Child vs. Adult" task significantly above the 80% stratified chance level. This confirms the hypothesis that the geometry of the covariance manifold captures more discriminative developmental information than simple linear correlation.

Reflective Analysis: While the MSDL method is effective, its performance is sensitive to the trade-off between subject-level fidelity (μ) and group-level sparsity (λ_1). The high variance in Partial

Correlation results highlights a limitation: isolating direct interactions in short fMRI time series remains statistically challenging (Fig 4). Overall, the results demonstrate that MSDL serves as a robust feature extractor for time-series analysis by balancing population consistency with individual variability.

4.2 Video Analysis: From Synthetic Validation to Traffic Atlases

In this section, we assess the adaptability of MSDL, originally designed to extract functional brain regions, to video feature extraction. The MSDL employed for this section is the optimized framework from the `nilearn` library, corresponding to Mensch et al. framework [2].

4.2.1 Validation and the Limits of Tracking

On **Dataset A**, the algorithm recovered ground truth maps with high fidelity, confirming that the implementation correctly minimizes the energy function when assumptions are met (reconstructed videos available in Colab). For **Dataset B** (Apparent Motion), MSDL successfully reconstructed the temporal sequence in U. Although the model lacks a causal tracking mechanism, it successfully reconstructed the 5 components and the "Ping-Pong" sequence in the extracted time series as we can see in Fig 5. Conversely, for **Dataset C** (Continuous Motion), the algorithm failed to recover the single component. Because MSDL models fixed spatial maps, it could not represent continuous translation of components. Instead, it produced a flip-book artifact, tiling the trajectory with multiple static components. Hence, high order models successfully reconstruct the component motion, but they interpret motion as a sequence of separate spatial components instead of a single entity with changing coordinates, making components uninterpretable. This confirms that MSDL is ill-suited for extracting or tracking objects that lack fixed spatial coordinates and temporal activations.

4.2.2 Extraction of Traffic Atlases from Real Data

Based on the theoretical mismatch identified in Dataset C, we hypothesized that MSDL would fail to track individual cars but would instead extract a "probabilistic atlas" of traffic flow. Applying MSDL to the BMC traffic dataset demonstrated that while the model fails at shape extraction for non-fluctuating shapes or moving shapes, it excels at unsupervised segmentation when assumptions are met. The cars, acting as non-stationary spikes, violated the MSDL assumptions. Consequently, the model forced these transient signals into the spatial maps. Due to the sparsity-inducing penalty, MSDL aggregated these transient movements into static, high-probability regions. Instead of isolating individual cars, the algorithm extracted two distinct spatial components corresponding to the two traffic lanes, as we can see in Fig 6. This result parallels the original paper's goal of defining a "functional parcellation" or population-level atlas. In the video context, MSDL effectively learned a "traffic atlas", segmenting the active road network from the static background.

References

[1] Gaël Varoquaux et al. "Multi-subject dictionary learning to segment an atlas of brain spontaneous activity". In: *Biennial International Conference on information processing in medical imaging*. Springer. 2011, pp. 562–573.

[2] Arthur Mensch, Gaël Varoquaux, and Bertrand Thirion. "Compressed online dictionary learning for fast resting-state fMRI decomposition". In: *2016 IEEE 13th International Symposium on Biomedical Imaging (ISBI)*. IEEE. 2016, pp. 1282–1285.

A Appendix

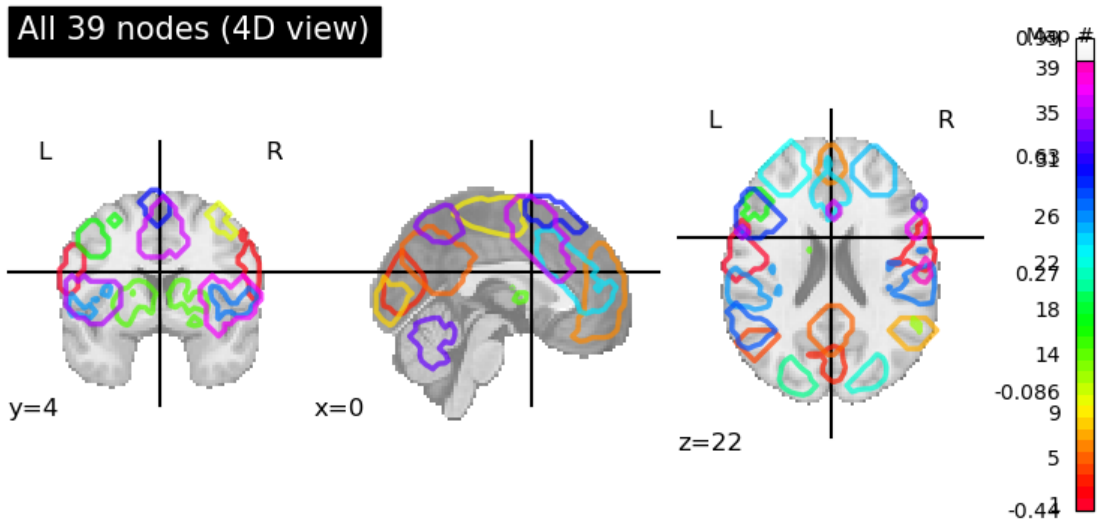

Figure 1: **The MSDL ROIs** The 39 nodes constitute a spatial atlas of the human brain that only emerges from unsupervised analysis of brain recordings.

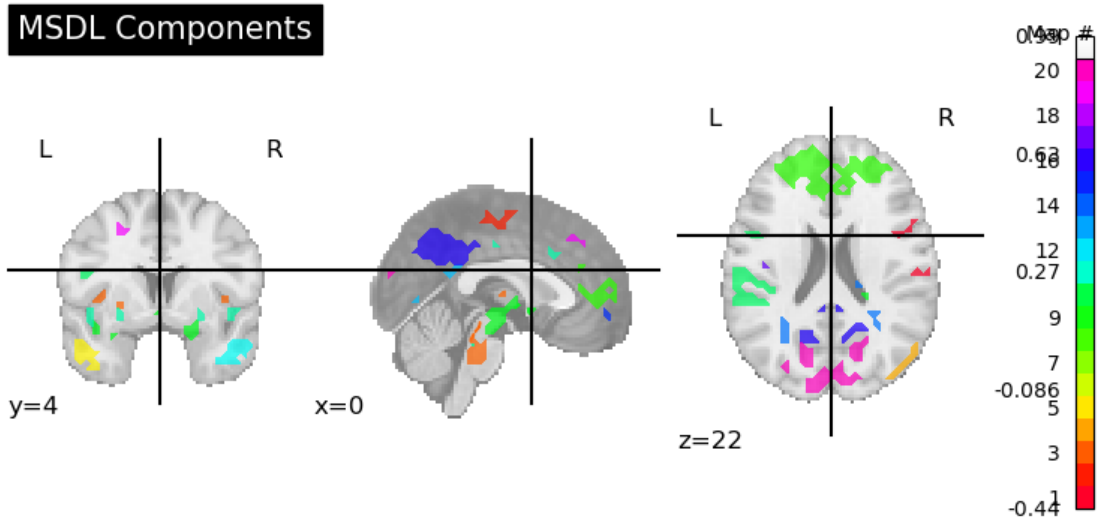

Figure 2: **Our custom components** Our implementation heavily relies on parameters and initialization, and is also much slower.

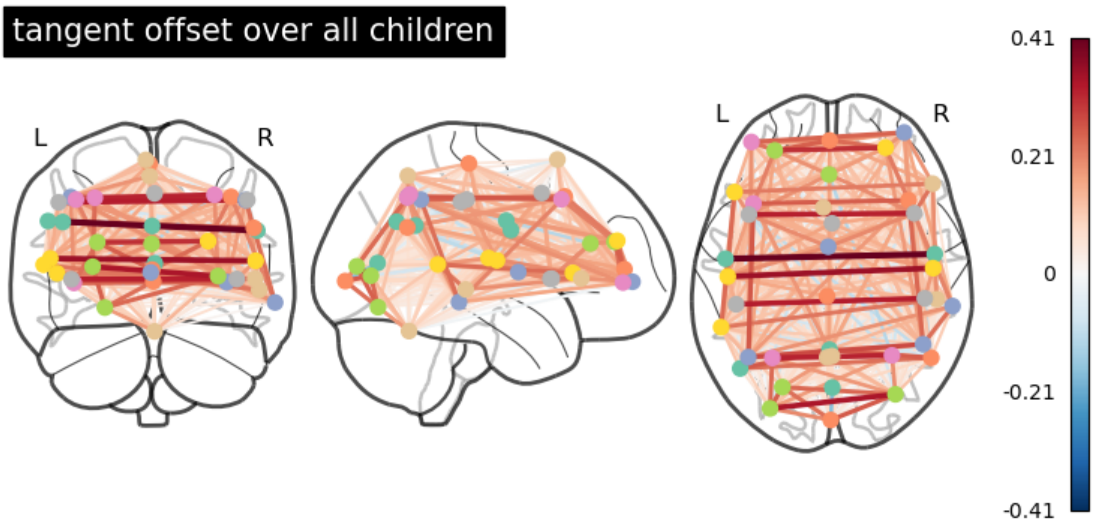

Figure 3: **Functional connectome using the MSDL atlas and tangent correlation measure** Those connectomes are emerging representations of spatial connectivity across the brain.

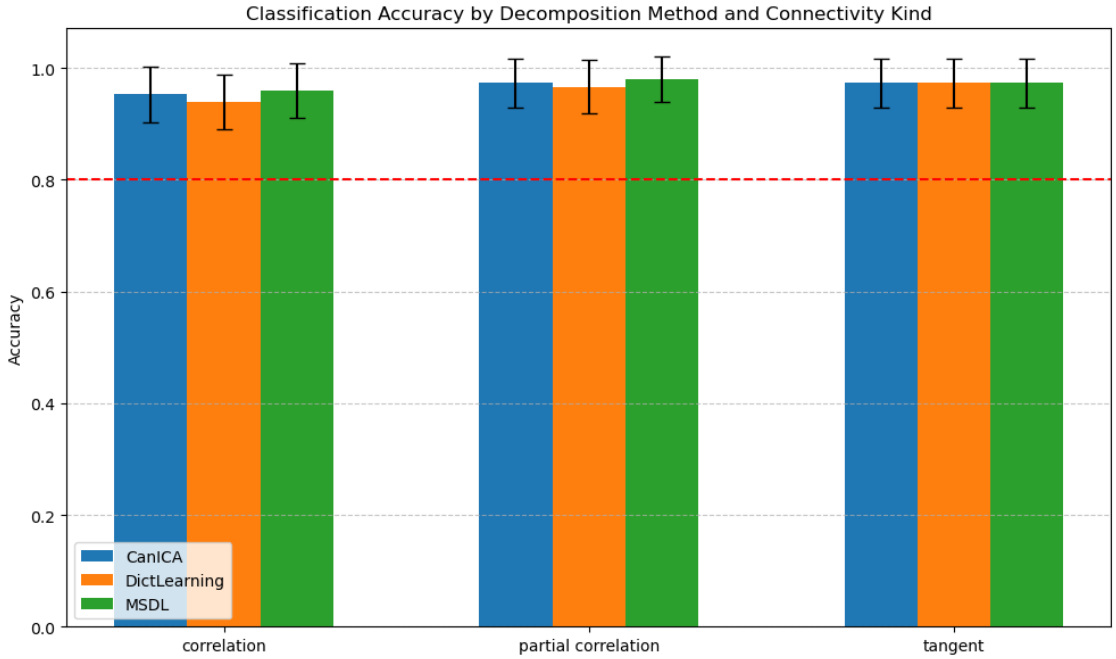

Figure 4: **Comparison of classification performance from the connectomes** Those functional connectomes can be used as features to infer age group with significant performance. However those benchmarks can have very different outputs with changes in parameters and dataset size.

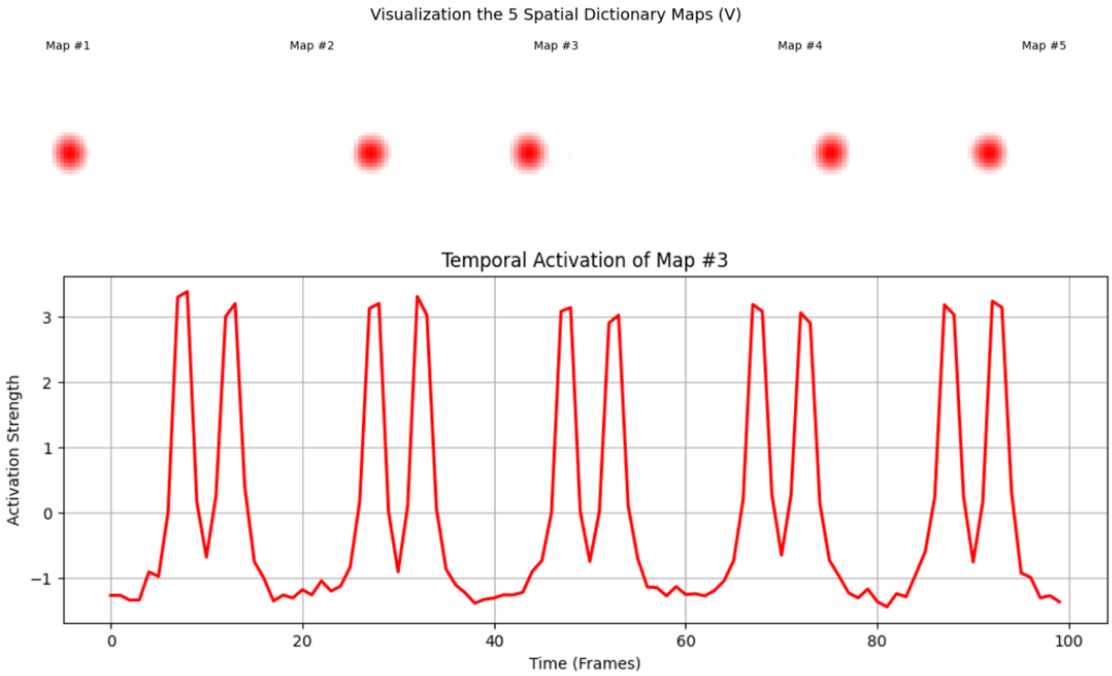

Figure 5: **Recovery of Apparent Motion Structure.** Top: Visualization of the 5 spatial dictionary maps (V) recovered from Dataset B, showing distinct, non-overlapping Gaussian blobs. Bottom: The corresponding temporal activation (U) for Map #3, demonstrating the model's ability to correctly reconstruct the "Ping-Pong" activation sequence despite lacking a causal motion tracking mechanism. We observed a "flip-book" artifact, tiling the trajectory with multiple static components when increasing the order of the model.

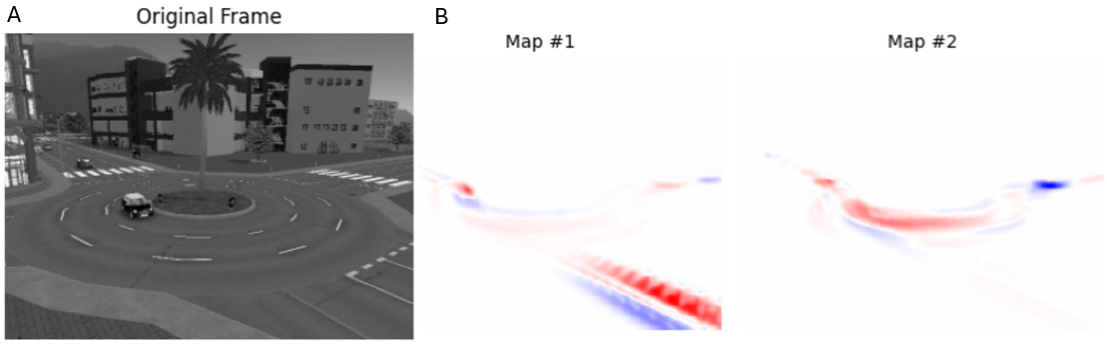

Figure 6: **Extraction of Traffic Atlases from Real-World Video.** (A) A representative frame from the BMC traffic dataset showing moving vehicles on a roundabout. The cars were going either frontward or going left at the roundabout. (B) The two spatial components (V) extracted by MSDL with a model order of $k = 2$. Instead of isolating individual cars, the model aggregates transient motion into static high-probability regions, effectively segmenting the active traffic lanes from the static background.